

Graduate Seminar

An introduction to stochastic calculus with applications to finance and SPDEs

Nicholas Eisenberg

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Stochastic Calculus

In Real Analysis we study measurable functions:

$$f : D \rightarrow R,$$

where both D and R are measurable spaces:

$$(D, \sigma(D), \mu_1) \quad \text{and} \quad (R, \sigma(R), \mu_2).$$

In undergraduate Calculus, we focus only on real valued measurable on $\mathbb{R}^d$:

$$f : D \subseteq \mathbb{R}^d \rightarrow \mathbb{R}^d,$$

and the measure space that is implied is the following:

$$(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda),$$

where λ is Lebesgue measure.

Stochastic Calculus

When studying Stochastic Calculus, and in general Probability Theory, we consider measurable functions,

$$X : \Omega \rightarrow \mathbb{R}^d,$$

and we consider the measure spaces

$$(\Omega, \mathcal{F}, \mathbb{P}) \quad \text{and} \quad (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda),$$

where Ω and $\mathcal{F}$ are referred to as the sample space and set of possible events and

$$\mathbb{P}(\Omega) = 1.$$

Definition (Random Variable)

A measurable function $X : \Omega \rightarrow \mathbb{R}^d$ is called a $\mathbb{R}^d$ -valued *Random Variable*. So essentially, Stochastic Calculus and Probability is just Real Analysis on Random Variables.

Stochastic Processes

Definition (Stochastic Process)

A stochastic process is a random variable that is indexed by set, T , which usually represents time:

$$X_t(\omega), \quad t \in T, \omega \in \Omega.$$

One can think a stochastic processes as a two parameter function

$$X_t(\omega) = X(t, \omega) : T \times \Omega \rightarrow \mathbb{R}^d.$$

Definition (Sample Path)

The variable ω represents the idea of a single possible outcome. The following function is then referred to as a *sample path*:

$$t \mapsto X_t(\omega).$$

Examples

Example

Suppose we leave a cup out in the rain. We may model this with a stochastic processes as

$$X_t(\omega) = \text{Amount of rain in the cup at time } t.$$

Since $X_t \geq 0$, maybe we could say that

$$X_t \sim \text{Gamma Distribution with } \mu := \mu(t) \text{ and constant variance } \sigma^2$$

If the rain is constant then it makes sense for the variance to be constant. Moreover, as time continuous, there will be more water in the cup on average, so the mean should be a function in time.

Brownian Motion.

Brownian motion is a very widely used stochastic processes. We will construct Brownian motion through a random walk.

Theorem

Consider a random walk $X = \{X_0 = 0, X_{t_1}, \dots, X_{t_n}\}$. Suppose that for $k \in \{1, \dots, n\}$ that

$$\mathbb{P}[X_{t_k} = -h] = \mathbb{P}[X_{t_k} = h] = 1/2.$$

and suppose that the time increments are all constant:

$$t_k - t_{k-1} = d \quad \forall k \in \{1, \dots, n\}.$$

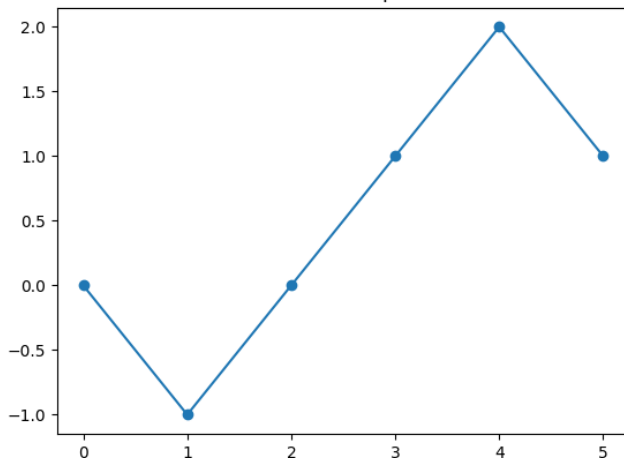
Suppose that $h^2 = d$. Now define

$$B_n(t_k) = \sum_{i=0}^k X_{t_i}.$$

Then B_n converges to a Brownian motion.

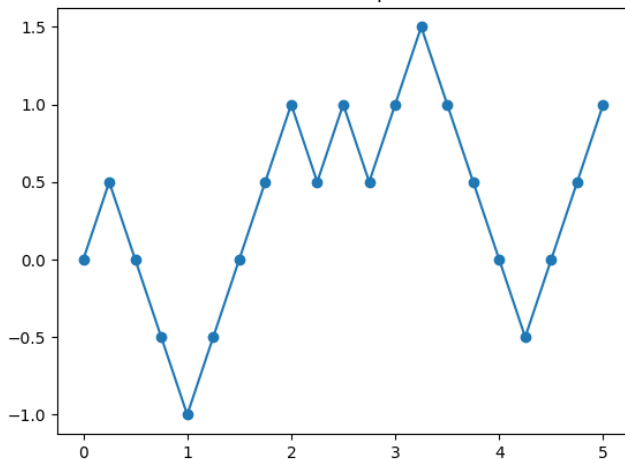
Random Walk Example

A random walk with jump size 1.0 and step size 1.0
number of steps = 5



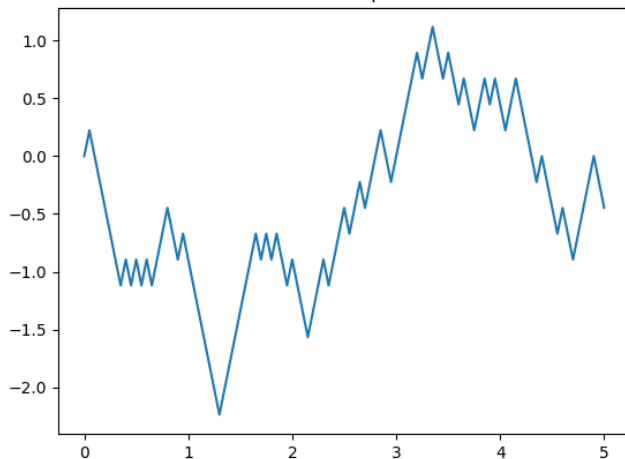
Random Walk Example

A random walk with jump size 0.5 and step size 0.25
number of steps = 20

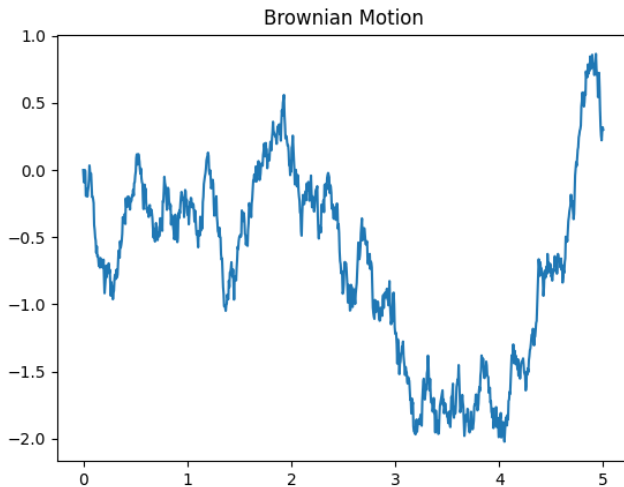


Random Walk Example

A random walk with jump size 0.224 and step size 0.05
number of steps = 100



Random Walk Example



References